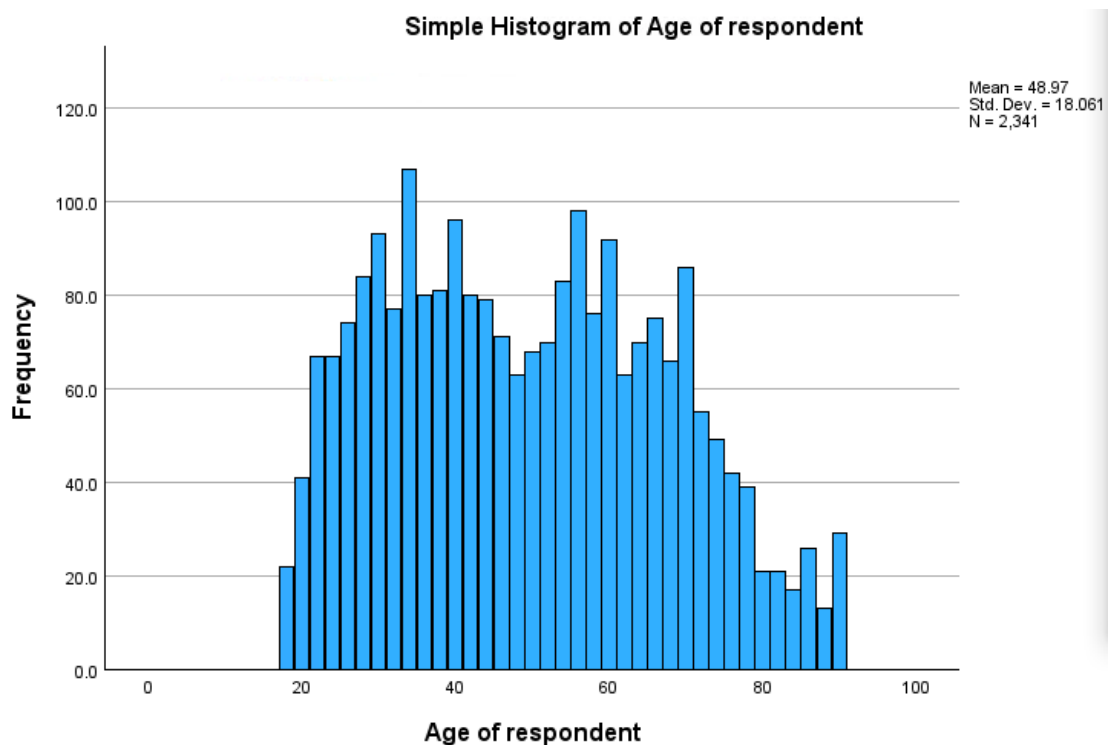


Part 1

Two continuous variables:

Descriptive Statistics						
	N	Minimum	Maximum	Mean	Std. Deviation	Variance
Age of respondent	2341	18	89	48.97	18.061	326.195
Number of children	2344	0	8	1.86	1.674	2.803
Valid N (listwise)	2338					

I selected age and number of children as the variables and used their mean values to measure the central tendency of the data. The average age of respondents is 48.97, and the average number of children is 1.86. The standard deviation for age (SD = 18.06) is relatively large, indicating a wide spread in respondents' ages. Meanwhile, the standard deviation of the number of children (SD = 1.674) is high relative to its mean (1.86), indicating more variability.



Statistics			
		Zscore: Age of respondent	Zscore: Number of children
N	Valid	2341	2344
	Missing	7	4
Mean		.0000000	.0000000
Median		-.0537836	.0863799
Mode		-.82894	-1.10816
Std. Deviation		1.0000000	1.0000000
Variance		1.000	1.000
Minimum		-1.71483	-1.10816
Maximum		2.21632	3.67000

The maximum z-score for the number of children is 3.67, which exceeds the common threshold of 3 standard deviations. This indicates that there is at least one outlier in the data.

Two categorical variables:

➔ Frequencies

Statistics			
		Belief in life after death	Confidence in business and industry
N	Valid	1129	1152
	Missing	1219	1196
Mean		1.70	2.96
Median		1.00	3.00
Mode		1	3
Std. Deviation		.980	.775
Variance		.960	.601
Minimum		1	1
Maximum		4	5

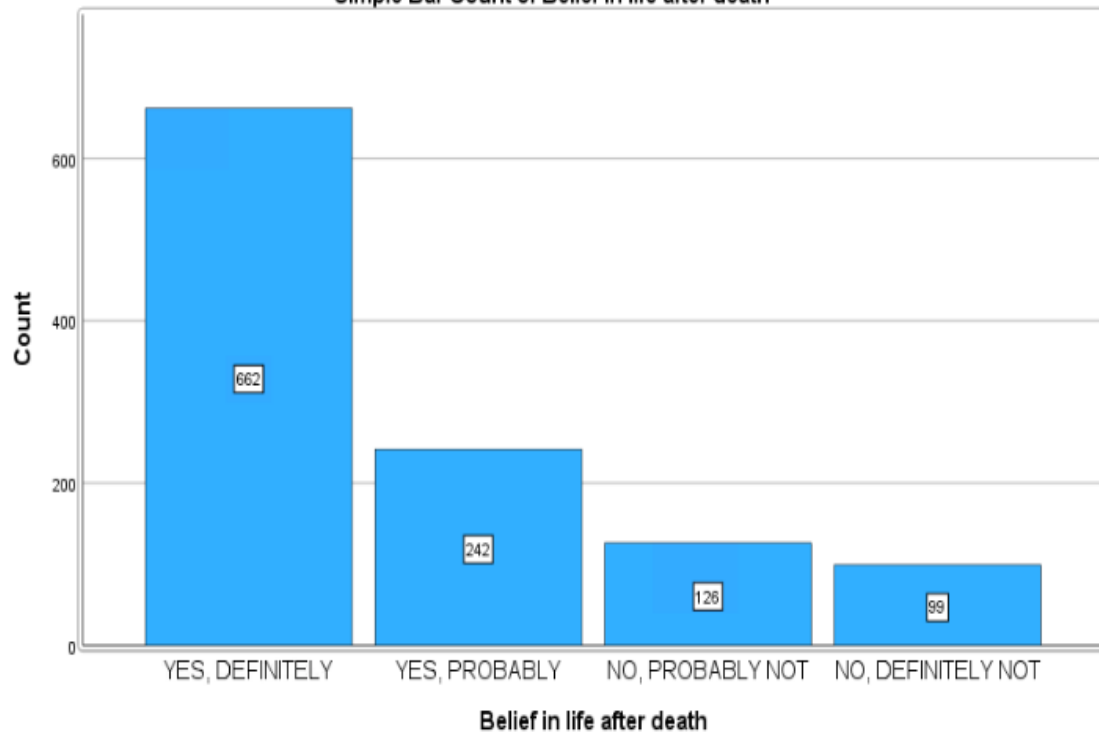
The central tendency of both variables is best described by the mode and median. For belief in life after death, both the mode and median are 1 ("Yes, definitely"). For confidence in business and industry, both are 3 ("some confidence"). The mean values are less meaningful because both variables are ordinal.

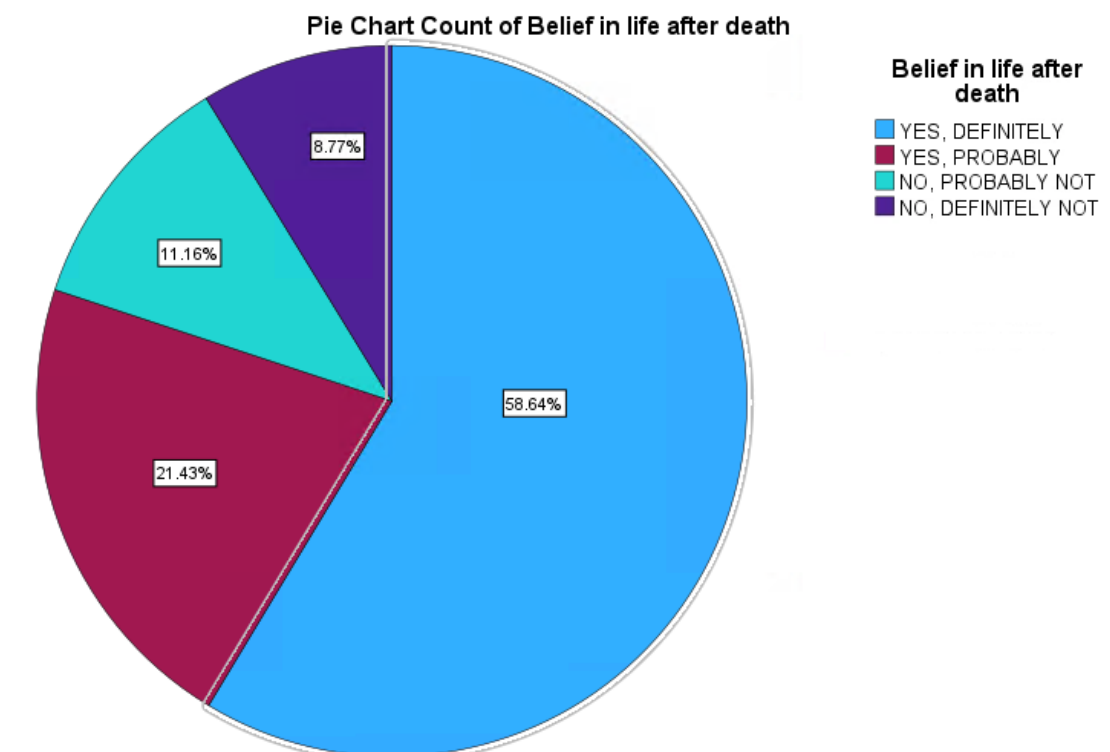
In terms of variability, belief in life after death has a higher standard deviation (0.98) than confidence in business and industry (0.775), indicating greater dispersion. However, because both variables are ordinal, the standard deviation should be interpreted with caution.

Belief in life after death

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	YES, DEFINITELY	662	28.2	58.6	58.6
	YES, PROBABLY	242	10.3	21.4	80.1
	NO, PROBABLY NOT	126	5.4	11.2	91.2
	NO, DEFINITELY NOT	99	4.2	8.8	100.0
	Total	1129	48.1	100.0	
Missing	IAP	1173	50.0		
	DK	36	1.5		
	NA	10	.4		
	Total	1219	51.9		
Total		2348	100.0		

Simple Bar Count of Belief in life after death





The frequency table shows that the majority of respondents believe in life after death. Specifically, 58.6% reported “Yes, definitely” and 21.4% reported “Yes, probably,” indicating that about 80.1% of valid respondents hold some level of belief. In contrast, only about 20% expressed disbelief.

However, it is important to note that a large proportion of responses (51.9%) are missing, which may affect the interpretation of the results.

Confidence in business and industry

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	COMPLETE CONFIDENCE	35	1.5	3.0	3.0
	A GREAT DEAL OF CONFIDENCE	223	9.5	19.4	22.4
	SOME CONFIDENCE	687	29.3	59.6	82.0
	VERY LITTLE CONFIDENCE	166	7.1	14.4	96.4
	NO CONFIDENCE AT ALL	41	1.7	3.6	100.0
	Total	1152	49.1	100.0	
Missing	IAP	1173	50.0		
	DK	15	.6		
	NA	8	.3		
	Total	1196	50.9		
Total		2348	100.0		

The majority of respondents reported “some confidence” in business and industry (59.6%), which is also reflected in the median and mode (both equal to 3). This indicates that responses are concentrated in the middle category.

However, a large proportion (50.94%) of responses are missing, which may affect the results.

Part 2

The purpose of this exercise was to use z-scores to standardize variables and identify outliers. The z-score is calculated by Subtract mean and divide by std deviation

$$Z = (X - M) / \sqrt{S^2}$$

Statistics			
		Zscore: Age of respondent	Zscore: Number of children
N	Valid	2341	2344
	Missing	7	4
Mean		.0000000	.0000000
Median		-.0537836	.0863799
Mode		-.82894	-1.10816
Std. Deviation		1.0000000	1.0000000
Variance		1.000	1.000
Minimum		-1.71483	-1.10816
Maximum		2.21632	3.67000

In this analysis, both variables were standardized, resulting in a mean of 0 and a standard deviation of 1. For age, the z-scores ranged from approximately -1.71 to 2.21, indicating that there were no extreme values beyond three standard deviations. However, for the number of children, the maximum z-score was 3.67, which exceeds the common threshold of 3 standard deviations and suggests the presence of at least one outlier. Overall, z-scores provide a useful way to compare values across variables and detect unusually large or small observations.